

Recovering Informative Estimates in Failed Placebo-Controlled List Experiments

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Abstract

List experiments are a popular technique to reduce sensitivity bias in surveys. Their widespread use has spurred considerable research design innovation. One suggestion to minimize bias from respondent non-strategic errors is to include a placebo statement in the control list. However, a poorly chosen placebo may introduce further bias, to a point in which estimates are no longer valid. This paper shows how to recover estimates in failed placebo-controlled list experiments. We present an example of a (double) list experiment on exposure to criminal governance tools in Uruguay in which the inclusion of a placebo item leads to invalid prevalence estimates. We then show how to leverage experimental assumptions and auxiliary information to recover informative estimates via partial identification bounds. A validation study confirms our bounds recover the true prevalence rate when it is known in advance.

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1 Introduction

List experiments are a popular technique to reduce sensitivity bias in surveys (Droitcour et al. 1991). Their widespread use in the discipline has spurred innovations aimed at establishing best research design practices (See Blair et al. 2020 for a review).¹

Some of these innovations seek to minimize *non-strategic* response errors. For example, inattentive respondents choosing alternatives at random or straight-lining may create an artificial correlation between treatment assignment and observed responses just because treatment lists are longer, which inflates prevalence rate estimates. In this case, the solution would be to include a zero-prevalence placebo statement in (at least some of) the control group so that list length is constant across experimental conditions (Riambau and Ostwald 2020; Agerberg and Tannenberg 2021).² Despite the promise of removing artificial inflation in a cost-efficient manner, placebo-controlled list experiments remain relatively infrequent in common social science applications (see Appendix A for details).

In this paper, we focus on the problem that emerges when the inclusion of a placebo statement backfires. In theory, one should include a placebo item in the control group that has (near) zero prevalence so that it does not induce further bias in the estimation of sensitive item prevalence rates. In practice, researchers may inadvertently choose a placebo item to which a non-trivial proportion of participants respond positively, either because the researcher underestimates its prevalence or because the placebo item stands out in a way that induces unintended responses. In this case, a failed placebo-controlled list experiment attenuates prevalence rate estimates.

This paper shows how to recover informative estimates in failed placebo-controlled list experiments using a partial identification approach. We first present a motivating example of

¹Another strand of the literature seeks to improve list experiments by developing new tools for estimation (e.g. Blair and Imai 2012; Blair et al. 2014), which is beyond the scope of this paper.

²Previous work also addresses *strategic* response errors. See Ahlquist (2017) and Blair et al. (2019) for a discussion.

a list experiment on the prevalence of exposure to criminal governance tools in Uruguay in which the inclusion of a placebo leads to prevalence rate estimates that are close to or less than zero. The multiple-sensitive item experimental design, plus additional statistical tests, suggests this attenuation of effects emerges from the inclusion of a failed placebo item over alternative explanations. We then show how to leverage list experimental assumptions and auxiliary information to produce informative estimate bounds. Since our motivating study is an extreme case, we apply our bounds to a validation study in which the prevalence rate of the item of interest is known in advance to show that our bounds approximate to the true proportion. We conclude with recommendations to safeguard against implementation issues in placebo-controlled list experiments.

2 A (Double) List Experiment with a Placebo Statement

2.1 Research Design

We conducted an online survey on a sample of 2,688 residents in the city of Montevideo, Uruguay in 2022.³ The main substantive goal of this survey was to document the prevalence of criminal governance actions in a context of high state capacity. Current research focuses mostly on cases characterized by more complex and violent versions of criminal governance, such as Mexico, Brazil, Colombia, or El Salvador (Magaloni et al. 2020; Arias 2017; Barnes 2021). Although at a smaller scale, criminal governance is still a persistent problem in other areas of the region, such as Argentina, Chile, and Uruguay (Auyero and Sobering 2019; Flom 2022; Fynn et al. 2024). Understanding the prevalence of criminal governance in these less

³Subjects were recruited via Facebook ads to balance costs with the goal of accessing respondents in areas where criminal governance is prevalent, wherein polling firms were wary of fielding in-person surveys. Pre-registration details can be found at <https://doi.org/10.17605/OSF.IO/TG6ZE>. The analyses presented here deviate from the pre-registration since we learned about the failed placebo control design after fielding the survey.

Table 1: Double list experiment baseline lists

List A	List B
Saw people doing sports	Saw people playing soccer
Visited friends	Chatted with friends
Attended activities by feminist groups	Attended activities by LGBTQ groups
Went to church	Went to charity events

Note: Respondents were asked to indicate how many things they have experienced in the last six months for each list. Baseline lists were presented in random order, items within lists were randomly shuffled. See Appendix D for complete question wording.

studied cases is crucial to determine whether mainstream theories apply beyond the original motivating cases.

We included list experiment questions in the survey. Participants were presented with two baseline lists of four things people may have experienced in the last six months and were instructed to indicate how many apply to them. Table 1 shows the two baseline lists, which were presented in random order. Appendix D shows the original questionnaire in Spanish and its translation to English.

We crafted baseline lists following standard advice in the literature to avoid ceiling and floor effects, which entails including at least two mutually exclusive items within each list and inducing positive correlation across lists (Glynn 2013). For example, we expected people who attended activities by feminist groups to be unlikely to also go to church (list A). Following the same line of advice, baseline lists are also constructed so that each item in list A has a pair in list B, which leads us to expect respondents to give similar numerical answers to both lists of baseline items. For example, seeing people doing sports (A) is paired with seeing people playing soccer (B). The purpose of these research design choices is to minimize the variance in estimates stemming from different latent responses to baseline lists. In other words, we want observed prevalence rates to be attributable to the sensitive item, not differences in the composition of baseline lists.

Table 2: Sample sizes by experimental condition

Item location		Sensitive item			
Sensitive	Placebo	Threaten	Evict	Donation	Work
List A	List B	328	322	339	341
List B	List A	340	339	326	333

Respondents were randomly assigned to see one of four sensitive items representing both negative and positive criminal governance actions:

1. Saw criminal groups threatening neighbors
2. Saw criminal groups evicting neighbors from their homes
3. Saw criminal groups making donations to neighbors
4. Saw criminal groups offering work to neighbors

In turn, the selected sensitive item would be randomly assigned to appear on List A or List B. Whichever list did not display the sensitive item would show the following placebo item:

- I did not drink *mate*

Mate is a popular herbal infusion in many South American countries, especially in Uruguay. We anticipated this would be a good placebo item as it is tantamount to asking in other contexts whether someone has not drunk coffee or tea in the last six months. Table 2 shows a breakdown of sample sizes by experimental condition.

We randomly assigned which sensitive item respondents see, the list in which it appears, and the order in which lists are presented. Based on self-reported neighborhood of residence within Montevideo, we formed two groups based on the police department’s classification of peripheral and non-peripheral zones. We conducted all randomization procedures independently within each block.⁴

⁴Respondents who did not provide a valid response to this question were dropped from the survey before random assignment to conditions. All results presenting treatment effects estimates are drawn from models that include block fixed effects.

Later in the survey, we asked about each sensitive item directly as a yes/no question within a longer battery of items. This lets us compare list experiment prevalence estimates with proportions estimated by direct questions. Crucially, we did not ask about the placebo item directly, nor we have access to reliable aggregate-level figures of its prevalence.

2.2 Results

As Table 2 suggests, we have what is equivalent to four independent double list experiments, or eight independent conventional “single” list experiments. We can also calculate the proportion of respondents who admitted to each sensitive item in a direct question. That means we can produce four different prevalence estimates per sensitive item (one direct-question based, three list-based).

Figure 1 shows results for all these quantities of interest. Each panel displays estimates for each sensitive item of interest, while each row indicates the estimator that corresponds to how the question was asked. Across items, the direct question suggest a positive, non-zero prevalence estimate that hovers around 10-20%. As usual, these estimates have much narrower confidence intervals relative to the list experiment questions, but if these are indeed sensitive questions, one would expect point estimates to under-report the true prevalence rate.

If the list experiment works as intended, one would expect it to yield prevalence estimates equal or higher than those drawn from a direct question. That is not the case in Figure 1: Point estimates are consistently around or below zero, in some cases with effect sizes implying -10% prevalence rates, which do not have a valid interpretation. In the context of a double list experiment, differences between single list estimates could hint at potential questionnaire design issues like results depending heavily on the baseline list (Diaz 2024), but in this case point estimates in both lists exhibit the same pattern, suggesting the issue extends beyond variation across baselines. Moreover, the pattern is consistent across both positive and negative forms of criminal governance, which suggests the problem goes beyond a particularly

problematic sensitive item.

We argue that the source of this unexpected pattern in estimates is the inclusion of the placebo item in the control lists. In the following sub-section, we discuss why the placebo statement backfired and present statistical tests to rule out alternative explanations.

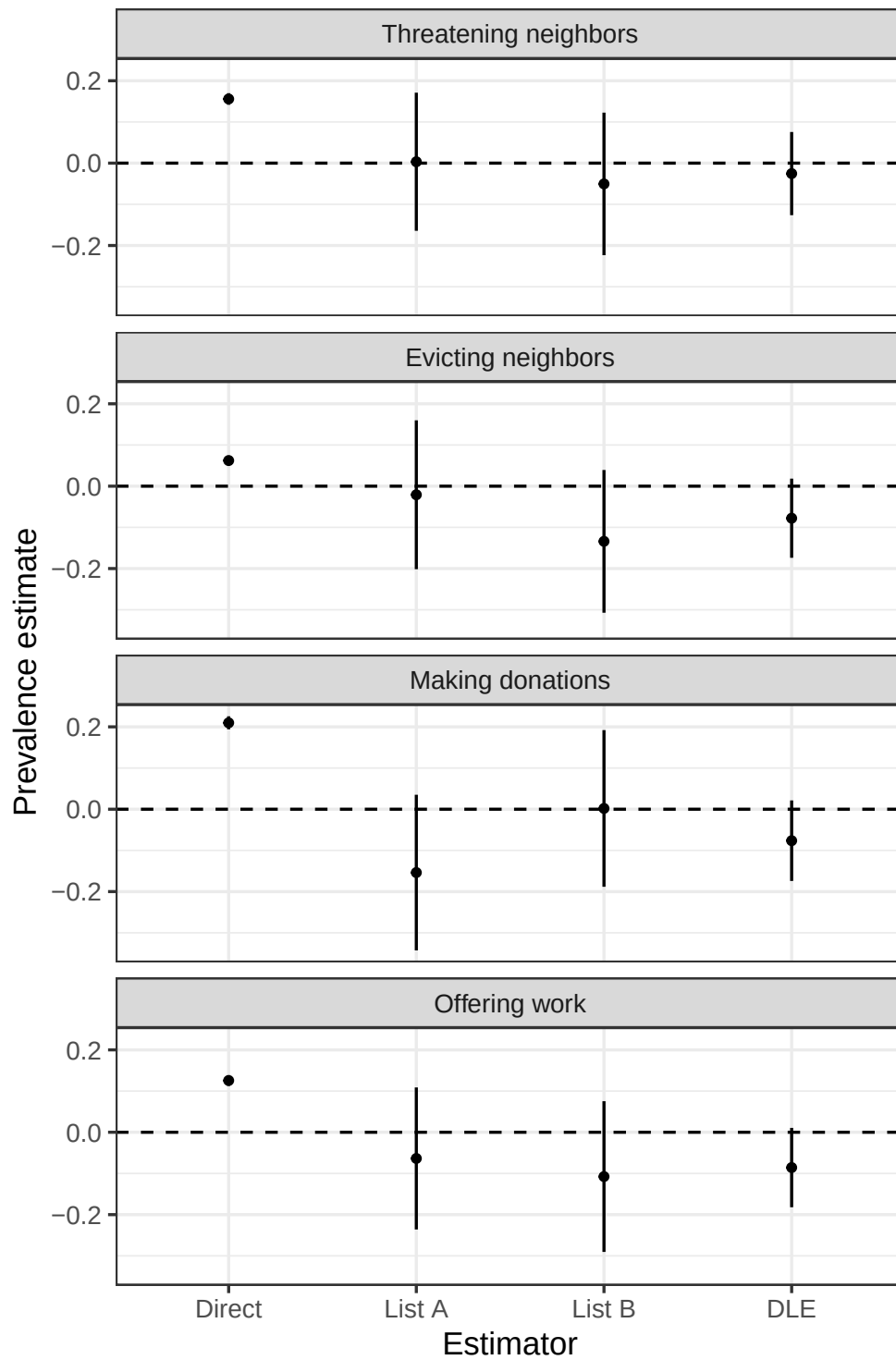


Figure 1: Prevalence estimates by sensitive item and estimator
Note: Vertical lines denote 95% confidence intervals.

2.3 Failed placebo control

Why did the placebo item backfire? The most straightforward explanation would be that the placebo statement does not have (near) zero prevalence, meaning it was chosen poorly. This is unlikely in our application since Uruguay is the highest per capita *mate* consumer (Gawron-Gzella et al. 2021) and anecdotal accounts suggest up to 85% of adults in the country consume it on a daily basis (which does not imply the remaining 15% did not drink within six months).⁵

An alternative explanation in the literature is that (near) zero-prevalence placebo items may still exhibit implicit non-zero responses in the list experiment question format because of respondent inattention. For example, Ahlquist et al. (2014) attribute estimated non-zero prevalence of alien abductions to respondents who rushed through the survey, thus committing non-strategic errors. The implication is that low attentiveness could produce a pattern like the one observed in Figure 1 because inattentive respondents produce low quality responses (e.g. straightlining, choosing a random number) that attenuate prevalence estimates toward zero.

To screen inattentive respondents, we included a pre-treatment mock vignette (Kane et al. 2023) that resembles a list experiment question. Respondents were asked to indicate how many of the following items apply to them:

1. I have been to the United States
2. I am a sports fan
3. I have a brother
4. I have more than one sister
5. I read the book “One Hundred Years of Solitude”
6. I voted by mail in the previous national election

⁵The daily consumption figure is not reliable. It comes from online searches in both English and Spanish that, as of writing, pointed primarily to listicles without clearly defined sources.

There was no random assignment of sensitive items, but the order of each item in the list was shuffled. Since voting by mail is not an option in Uruguay, respondents who indicate all six items apply to them are revealing they did not read the items carefully enough.⁶ About 8% of respondents (216/2688) indicated all six items apply. We find no evidence of these inadmissible responses changing prevalence estimates. Admittedly, this is a lower bound approximation since inattentive respondents may also choose other options. However, we also find no evidence of attention check responses affecting how long respondents spend on the list experiment questions (see Appendix C for details and additional tests).

Another explanation in the literature points to *localized* inattention. Riambau and Ostwald (2020) and Agerberg and Tannenberg (2021) caution against using items that are non-zero because they are implausible in the context of the baseline list (e.g. I can reach the moon by jumping), as it would translate into a perception that the list experiment question is unserious or was poorly designed, which also lowers response quality as it leads respondents to not pay close attention to the items.

Our double list experiment setting provides an opportunity to test for this pattern. If it is true that the placebo item makes respondents think the question is not serious, then presenting the placebo item in the first list would also lead to altered responses in the second list in a pattern that would push prevalence estimates closer to zero relative to respondents that saw the placebo item in the second list.

In other words, if placebo items decrease response quality in list experiment questions, then one would expect them to induce carryover design effects when presented in the first list. Table 3 tests for carryover design effects using the difference-in-differences tests proposed in Diaz (2024), except that the treatment indicator is flipped so that the list with the placebo item takes a value of one and is zero otherwise.⁷

⁶The question wording in Spanish strongly implies, but does not mention directly, that the vote by mail refers to the previous presidential election in Uruguay by referring to it as a “national” election.

⁷The only consequence of this decision is a sign flip on the test statistic. The absolute value statistic does not change.

Table 3: Testing for carryover design effects from placebo item

Item	Statistic	p-value	n
Threatening neighbors	-0.116	0.407	671
Evicting neighbors	-0.085	0.576	664
Making donations	0.236	0.155	668
Offering work	0.111	0.468	677

Note: Statistic represents change in double list prevalence estimates when placebo item appears in the first list. p-values drawn from a model that includes block fixed-effects and clustered standard errors by respondent.

Test statistic values suggest double list prevalence estimates tend to be smaller when the placebo item is presented in the first list for the sensitive items that represent negative criminal governance tools, but positive for items that represent positive tools. However, the p-values suggest we do not have enough evidence against the null of indistinguishable effects, which leads us to interpret the patterns in Figure 1 do not stem from localized inattention.

Even if inattention is not the primary driver of the pattern in Figure 1, respondents may still actively select the placebo item if it stands out. For example, our placebo item may attract respondents’ attention since it is the only one that is worded negatively. Since the actual experiment appears shortly after mock vignettes that test respondents with questions like “how many items did you see in the previous list”, some participants may be led to think this is some sort of desirable or correct answer that they need to signal association with by providing higher numerical responses.

Alternatively, because all other items are positive statements and the only thing that makes our placebo item negative is the presence of a leading “no” in the original Spanish wording (see Appendix D), some respondents may interpret the item as a positive statement as well, producing responses opposite to what we intended.

Either way, the point is that respondents who are not egregiously inattentive can still count

the placebo item in their response even if it does not apply to them. Aronow et al. (2015) propose a statistical test that helps screening for such response pattern in the aggregate. The test statistic is the list experiment prevalence estimate on the subset of respondents who have admitted to the sensitive item on the direct question. In the conventional design, if the assumptions of no liars, no design effects, and monotonicity hold, then the statistic should be statistically indistinguishable from one. This implies the list experiment works as intended by accurately capturing those who openly admit to the sensitive item.

In our design, this statistic can be different from one if all the list experimental assumptions hold, but the placebo item exhibits non-zero prevalence, if participants count the placebo item in their response.⁸ Table 4 shows the results for this test, applied to every sensitive item and prevalence estimator possible under our design. Test statistics are consistently negative, and p-values provide sufficient evidence to reject the null of the prevalence estimate being equal to one, with the exception of one case in which the sample size is too small (List A, Evicting neighbors).

The pattern of test results observed in Table 4 suggests that average responses to baseline lists including the placebo item are greater than in baseline lists that include the corresponding sensitive item. This is possible under two circumstances. First, and the most likely explanation in this case, respondents are actively selecting the placebo item across the different sensitive items. This can be considered a form of strategic error.

Second, respondents are actively deflating responses in lists that include the sensitive item, which entails a violation of the no design effects assumption. There are three reasons to believe this is not the case: (1) The negative test statistic appears across sensitive items, whereas design effects often vary depending on the sensitive items (e.g. Diaz 2024); (2) A violation of the “no design effects” assumption of a magnitude sufficient to produce negative

⁸Hatz and Randahl (2024) discuss another case in which this statistical test yields non-null results even if conventional assumptions are not violated. If sensitivity bias operates in different directions across subgroups, then the test statistic can differ from one.

Table 4: Testing for non-zero placebo prevalence

Item	Statistic	p-value	n
<i>List A</i>			
Threatening neighbors	-0.851	0.000	102
Evicting neighbors	-0.585	0.235	32
Making donations	-0.539	0.021	133
Offering work	-0.820	0.011	64
<i>List B</i>			
Threatening neighbors	-0.612	0.010	102
Evicting neighbors	-1.106	0.003	32
Making donations	-1.259	0.000	133
Offering work	-1.131	0.004	64
<i>Double</i>			
Threatening neighbors	-0.755	0.000	102
Evicting neighbors	-0.860	0.000	32
Making donations	-0.916	0.000	133
Offering work	-0.971	0.000	64

Note: Statistic represents prevalence estimate among those who admit to sensitive item in direct question. p-values drawn from a model that includes block fixed-effects and clustered standard errors by respondent, testing the hypothesis of Statistic = 1.

test statistics would have been captured in the test for carryover design effects in Table 3; (3) descriptive statistics and χ^2 tests in Appendix B do not provide sufficient evidence for such a distributional shift.

To summarize the results in this section. The prevalence estimates in Figure 1 suggest the inclusion of placebo items introduced considerable attenuation bias across all sensitive items. Statistical tests suggest this is not primarily driven by inattention, but by respondents actively choosing the placebo item, either because it applies to them or, most likely, because it stands out in a way the leads respondents to count it in their response even if it does not apply.

As a note of caution, the tests presented here only highlight what happens in average. There-

fore, not having enough evidence to reject null hypotheses does not imply that inattention, general or localized, is not present. Similarly, just because we expect non-drinkers to be rare does not imply that they are not present in our data. The important lesson from this section is that the researcher needs to be convinced that an unexpected pattern of estimates is driven by the inclusion of the placebo item, as opposed to mere inattention. This step is crucial, because the refinements we propose in the next section depend on the researchers believing the standard requisite list experiment assumptions hold.

3 Recovering Informative Estimates

If one can claim that the presence of a non-zero prevalence placebo is the primary source of the pattern of prevalence estimates in Figure 1, then the task is to reconstruct how responses would have looked like had the placebo item not been included in control lists. If the data allowed the researcher to know exactly which respondents the placebo applies to, then one could simply correct the biased prevalence estimates by subtracting one from the control responses of participants for which the placebo item applies.

Because we did not include a direct question about the placebo item, we do know not exactly which responses need to change. This implies more than one valid prevalence estimate can be obtained by “cleaning” the responses contaminated by the placebo item. In other words, the estimand is no longer point-identified, but one can still compute bounds around it. The advantage of computing non-parametric sharp bounds is that they do not assume anything beyond the observed data and knowledge of how the experiment was conducted. The disadvantage, as it is usually the case with non-parametric partial identification of experimental treatment effects, is that sharp bounds tend to cover a wide range and are therefore substantively uninformative (Aronow and Miller 2019). In our case, the list experiment setting provides additional structure in the form of required assumptions and auxiliary information that enables the computation of more informative bounds.

3.1 Assumptions

The goal in nonparametric partial identification is to identify the narrowest possible bounds using as little information as possible. In experiments, sharp bounds are derived exclusively by the observed data and standard experimental assumptions (Manski 1990). In this context, the non-interference part of SUTVA implies that only the responses in which the placebo item was included would be affected by its removal. One can compute lower and upper values of these sharp bounds by imputing the maximum (4, since that is the “updated” number of items) and the minimum (0) value in the control responses, while leaving treated responses unchanged.⁹

The left column of Figure 2 shows the resulting sharp bounds computed in this manner, with the corresponding 95% confidence intervals based on 1,000 bootstrap replicates. As expected, the possible values far exceed the logical range of prevalence estimates, the lower bound even reaches up to a prevalence of about -2 across sensitive items.

How can one compute more informative estimates? The standard list experiment assumptions provide additional structure on the possible values that responses would take in the absence of a placebo item. The “no liars” assumption requires respondents to not actively count the sensitive item when presented to them in the list if it does not apply to them. The “no design effect” assumption requires implicit responses to baseline items to not be altered by the inclusion of a sensitive item. These assumptions are required for prevalence estimates to have a valid interpretation (see Blair and Imai 2012 for details).

These assumptions are explicitly made about the sensitive item, but the “no design effects” assumption is implicitly extended to the placebo item in the literature proposing placebo-controlled designs (Agerberg and Tannenberg 2021; Riambau and Ostwald 2020). A placebo-controlled list experiment only works if one expects responses to baseline lists to not be altered by the the inclusion of a placebo item in it.

⁹The reversal is intentional. If only control responses are altered, then imputing the maximum value yields the smallest possible treatment effect, and viceversa.

Extending the “no design effects” assumption to the placebo item has two implications when computing bounds: (1) Responses to control lists can only decrease (2) by a magnitude of one (unless the observed response is already zero). The lower bound in this context is calculated by decreasing all observed control responses that already have a maximum value when the placebo is included (5). This is equivalent to making the conservative assumption of only decreasing by one the responses of those who unequivocally reveal holding the placebo item. The upper bound is then obtained by decreasing all observed control responses by one unit, which would be equivalent to assuming everyone holds the placebo item. This upper bound is unrealistic, but the best one can do without bringing additional assumptions or information about the distribution of the placebo item.

The right column in Figure 2 shows the bounds that follow from incorporating list experimental assumptions. These now approximate the logical $[0, 1]$ range of a valid prevalence estimate, but are still uninformative in substantive terms.

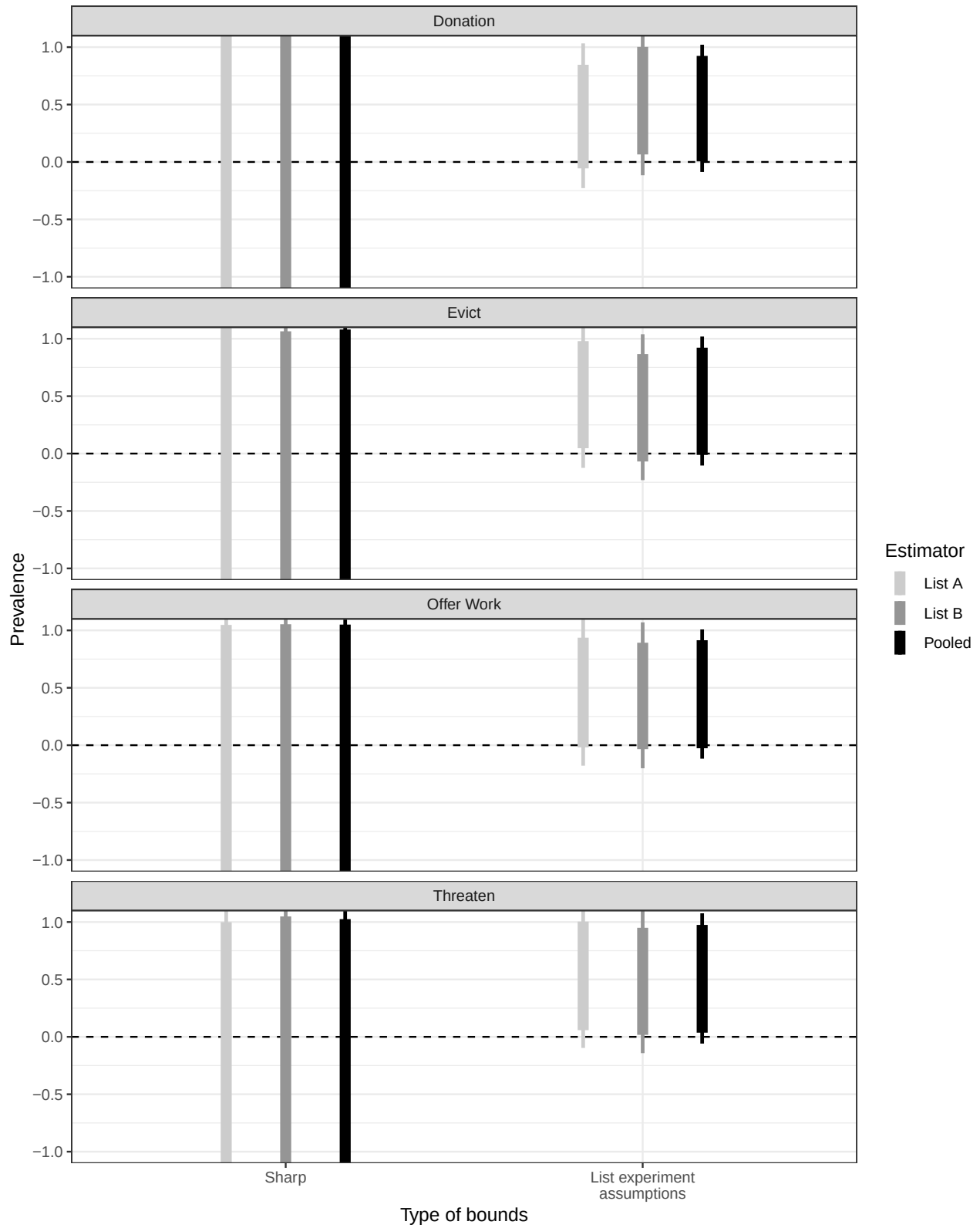


Figure 2: Nonparametric bounds by sensitive item

Note: Thinner lines indicate bootstrapped 95% confidence intervals. Zoomed in to facilitate visualization.

3.2 Known placebo proportions

The bounds computed under the list experimental assumptions depict extreme cases of how control list responses would have been altered in the absence of a placebo item. The lower bound, derived by decreasing already maximum values by one unit, has the desirable property of assuming only what one can unequivocally observe in the data. However, the upper bound implies a placebo item that applies to every single unit, which is far from the ideal zero prevalence target.

One way to impose a more realistic upper bound would be to restrict the proportion of units with less than maximum control responses that are allowed to decrease their answer by one unit. If the researcher knows the proportion of placebo item holders in the target population from auxiliary information, they could impose a lower ceiling on the upper bound.

The new upper bound is then given by

$$\hat{\tau}_H = (1 - \delta) \times \bar{V}_{obs} + \delta \times \bar{V}_{max} \quad (1)$$

where δ is the proportion of known placebos, $\bar{V}_{obs}$ is the list experiment estimate based on observed data, and $\bar{V}_{max}$ is the list experiment estimate based on the hypothetical vector of observations in which every control response is reduced by one unit, the previously discussed unrealistic upper bound. This equivalent to the standard approach to estimate partial identification bounds by computing a weighted average, which is numerically the same as finding the vectors that yield the most extreme estimates manually or automatically (Aronow and Miller 2019).

Figure 3 shows the resulting estimates with this new upper bound. Since we did not ask directly about the placebo item, nor we have access to reliable information from auxiliary data, we simulate estimates at discrete values in the $[0.05, 0.3]$ range. Overall, this produces much more informative upper bounds, especially at ranges that imply a somewhat useful

placebo item with small yet non-zero prevalence (e.g. 0.05, 0.1). However, the problem persists in the lower bound, as estimates still exceed the range of admissible values for a prevalence estimate measured as a proportion.

Notice that if one were to ask a direct question yes/no about the placebo item, and be willing to claim it is measured with negligible error, then one would be able to exactly identify which responses should decrease by one unit, in which case the bounds would collapse into a point estimate again. If one asks directly about the placebo item but believes these are measured with non-negligible error, then the bounds proposed here may still be useful to assess the sensitivity of estimates to potential measurement issues in placebo item proportions drawn from a direct question.

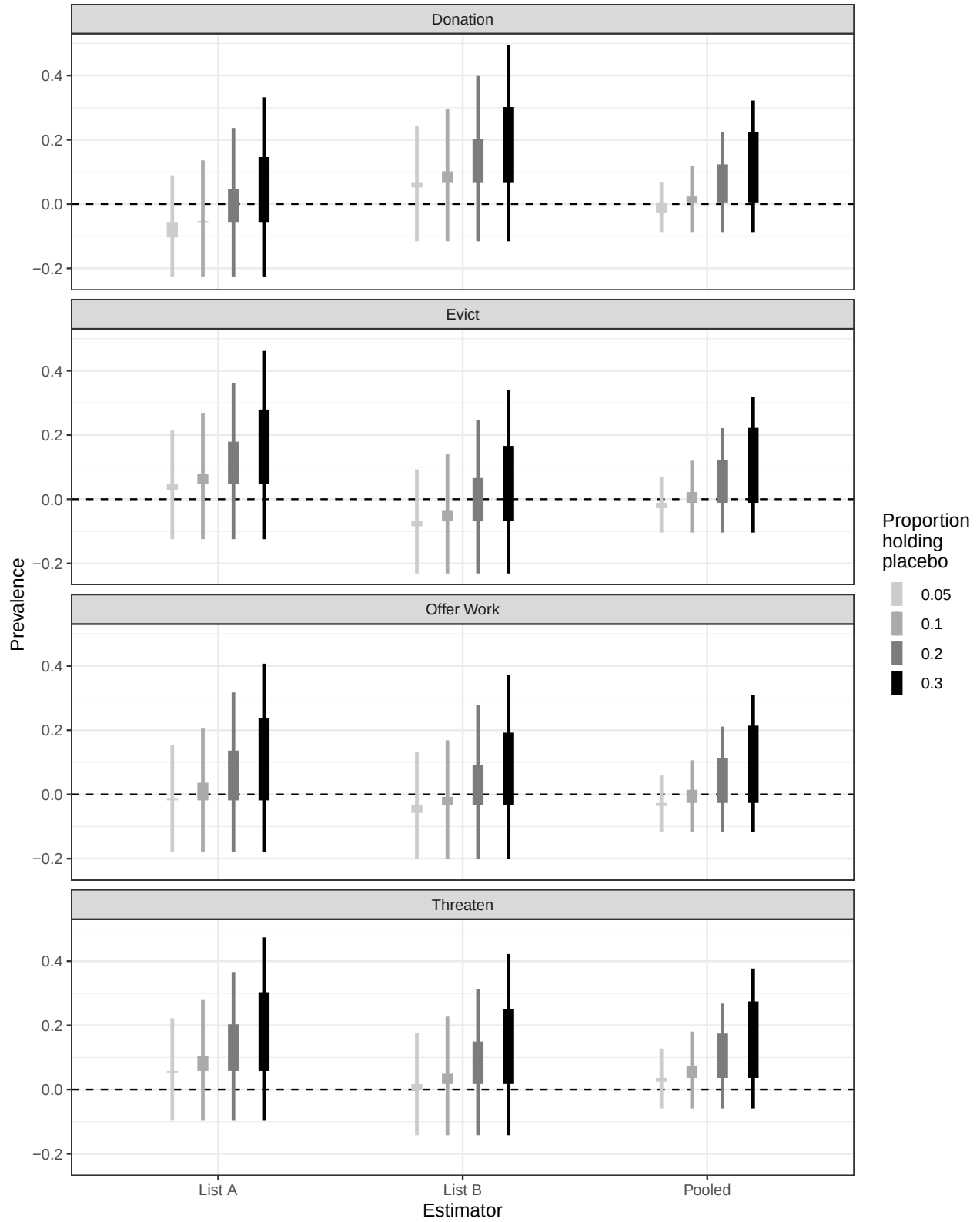


Figure 3: Nonparametric bounds by different assumed placebo proportions
Note: Thinner lines indicate bootstrapped 95% confidence intervals.

3.3 Incorporating direct questions

A general limitation of list experiments is that they produce wider confidence intervals relative to alternative approaches to sensitive survey questions (Rosenfeld et al. 2015; Blair et al. 2020). As our findings so far suggest, the challenge of low statistical precision is even more pronounced when computing partial identification bounds.

Previous work aims to alleviate the standing of list experiments on the bias-variance frontier by proposing alternative methodologies that improve statistical precision (e.g. Aronow et al. 2015; Chou et al. 2017; Lu and Traunmüller 2026). Aronow et al. (2015) propose an estimation procedure that combines list experimental estimates with direct questions. The idea is that one does not need an indirect questioning technique for respondents who openly admit to the sensitive item in a direct question, so their response to the list experiment questions can be “replaced.” This informs the combined prevalence estimator

$$\hat{\tau}_{comb} = \bar{Y} + (1 - \bar{Y})(\bar{V}_{Y_i=0}^*) \quad (2)$$

where $\bar{Y}$ is the proportion of respondents in the sample who admit to the sensitive item in the direct question and $\bar{V}_{Y_i=0}^*$ is a list experiment estimate among respondents who do not admit (see Aronow et al. 2015 for details).

In a standard application, $\bar{V}_{Y_i=0}^*$ would correspond to the observed list experimental estimate $\bar{V}_{obs}$. In our context, one could produce narrower bounds than those presented on the right column of Figure 2 by replacing $\bar{V}^*$ with the list experiment estimate based on the vector of observations in which every control response is reduced by one unit $\bar{V}_{max}$ to produce the upper bound, and the estimate based vector of observations in which only those with maximum values decrease their response by one unit, call it $\bar{V}_{min}$.

Figure 4 shows the bounds produced by the combined estimator, along with their list experiment analogue to facilitate comparison. Figure 5 shows bounds combining the known

placebo proportion refinement with estimators that combine list experiment with direct questions. Regardless of figure, incorporating direct questions in the estimation procedure leads to narrower bounds across all sensitive items.

The increase in precision is proportional to the share of respondents admitting to the sensitive item in the direct question. For example, the most pronounced increase in precision appears in the first panel of Figure 4, which depicts bounds for the sensitive item of seeing criminal organizations making donations to neighbors. In the direct question, about 21% of respondents admitted to this item. In contrast, the least pronounced change in bounds appears in the second panel, seeing criminal organizations evict neighbors from their homes, for which only about 5% of respondents admitted to in the direct question.

The main draw of using the direct question combined estimator is that it constitutes a design-based approach to narrow bounds. This is because the estimator itself is more precise, so the gains occur *before* computing the bounds. Similar improvements may be obtained with other estimators that improve precision by relying on multivariate regression (Imai 2011; Glynn 2013) or auxiliary administrative information (Chou et al. 2017).

Another advantage of using a more precise estimator to compute bounds is that it provides a principled way to narrow the lower bound. As Figure 3 shows, knowing the proportion of placebo holders in the population of interest can decrease the upper bound substantially, but it does not move the lower bound. The implication is then that both strategies should be considered as complementary: Before relying on auxiliary information to refine bounds, one should make sure to use the most precise estimator available.

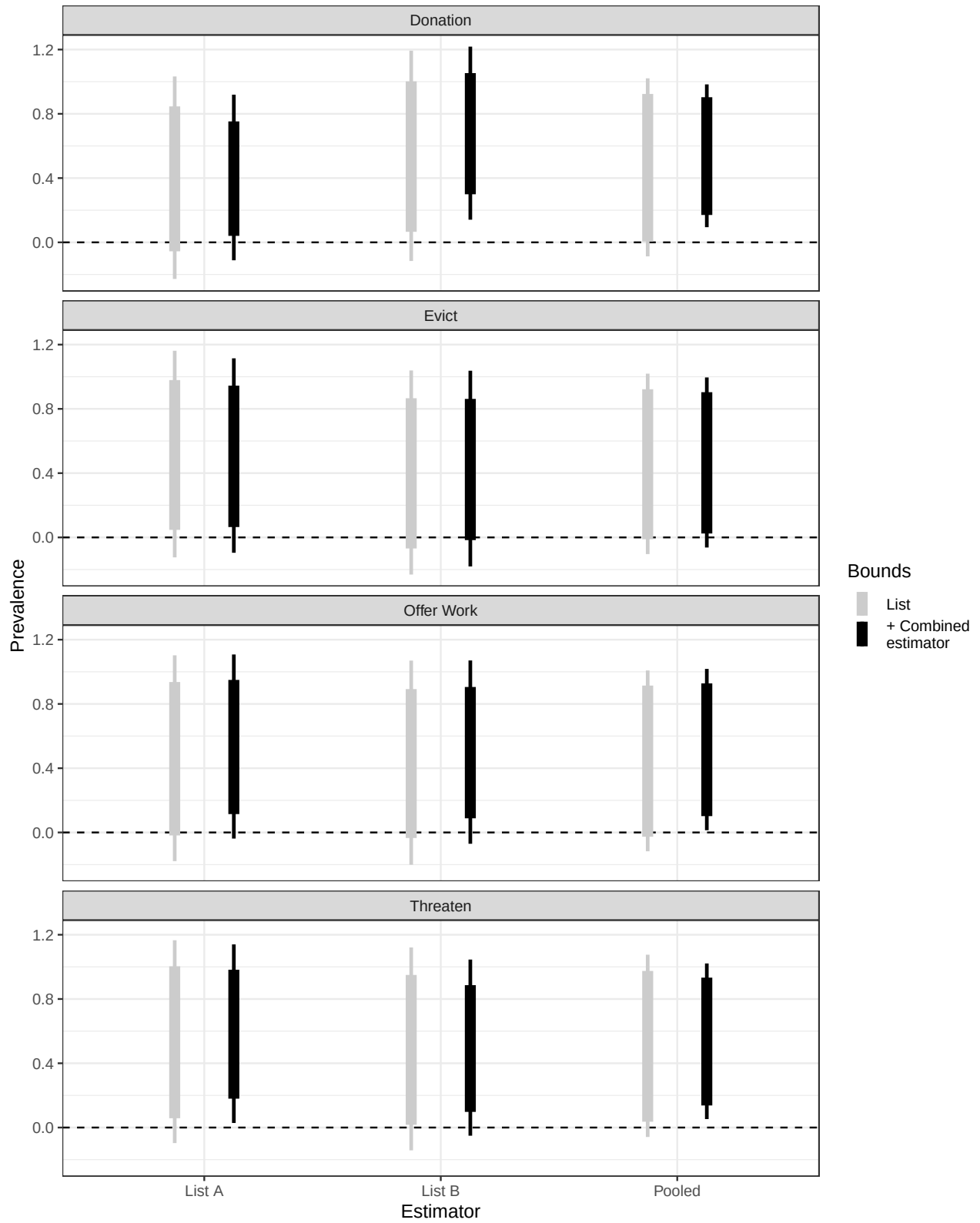


Figure 4: Comparing bounds using the standard list experimental estimator and the direct question combined estimator

Note: Thinner lines indicate bootstrapped 95% confidence intervals.

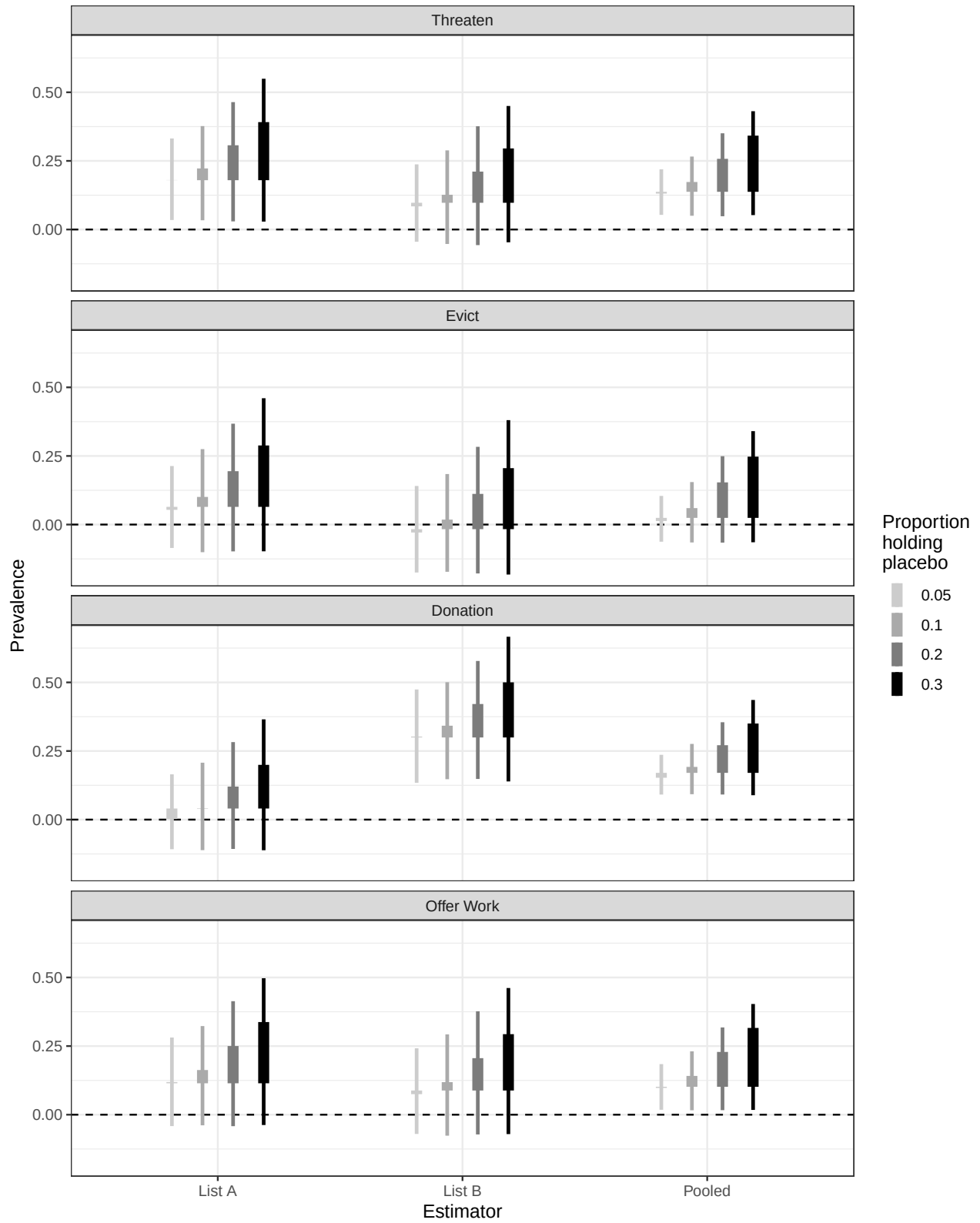


Figure 5: Bounds combining known placebo proportion and the direct questions by estimator type

Note: Thinner lines indicate bootstrapped 95% confidence intervals.

4 A more realistic validation study

Our running example on the prevalence of criminal governance tools in Uruguay is illustrative of the challenge of failed placebo-controlled list experiments. However, it is an extreme case and a poor reflection of what researchers may typically encounter when they implement this design. In this section, we show a more realistic application of our bounds using data from a previously published validation study on placebo-controlled designs.

Agerberg and Tannenbergs (2021) conduct an online (single) list experiment on a sample of 4,973 participants in China via Lucid.¹⁰ Respondents were prompted to answer how many items apply to them. The baseline list included:

- The government is the employee of the people and should do as the people wish.
- The government is like a parent and should tell the people what to do.
- I live in the same city where I was born.
- Generally speaking, most people cannot be trusted.

The goal of the original study was to show how placebo-controlled list experiments do not necessarily eliminate the bias that emerges from different list lengths in the conventional design, and that it may even introduce bias in the opposite direction (i.e. deflate rather than inflate estimates). Therefore, the item of interest was not necessarily sensitive, but one with known population proportion:

- I was born in the year of [BLANK] or the year of [BLANK].

The blanks were filled by randomly sampling from a list of six possible combinations of zodiac animals, which map to birth years and are common knowledge in China. For example, one option included the year of the Rat or the Ox, another the year of the Dragon or the Snake. Each one of the six possible items has a non-overlapping pair of zodiac animals, distributed so that the expected prevalence rate is known in advance to be 1/6 (see Agerberg

¹⁰Replication materials were obtained from Tannenbergs (2021).

and Tannenberg 2021 for details).

On top of a conventional treatment and control group, respondents were also be assigned to a placebo-control condition that included the baseline list plus the following placebo item:

- I was born in the 2000s or 1970s

The actual decade was selected based on the respondent’s self-reported age earlier in the survey so that it would be impossible (or illogical) to respond positively to this item.

One third of the respondents were assigned to the control, placebo, and treatment group. The main advantage of this design is that we can compare prevalence estimates using the control and placebo group as reference for the treatment group separately, which leads to different point estimates.

Another advantage is that we can use the difference in means between the placebo and the control group as a proxy for the proportion of placebo holders in the population of interest. Following this approach, our estimate is that about 17.4% of respondents “hold” the placebo item in the sense that they selected it even if it does not apply to them. We can then use this figure as our known placebo proportion as discussed in Section 3.2.¹¹

The first two columns of Figure 6 show point estimates using either the conventional control or the placebo group as a reference for the treatment group that includes the item of interest. These results match exactly what is reported in Agerberg and Tannenberg (2021): Both conventional and placebo control lists produce estimates that deviate from the true value in this application, which the authors use as a motivation to argue for mixed-control lists in which only some respondents in the control group see the placebo item.

The third column of Figure 6 shows the bounds computed on the contrast between the treatment and placebo group, drawing on the list experimental assumptions discussed in Section 3.1 and the known placebo proportion approach from Section 3.2 to further refine

¹¹In practice, researchers may have access to such a figure from auxiliary sources like census figures or data from nationally representative surveys. The figure here is used for illustration purposes only.

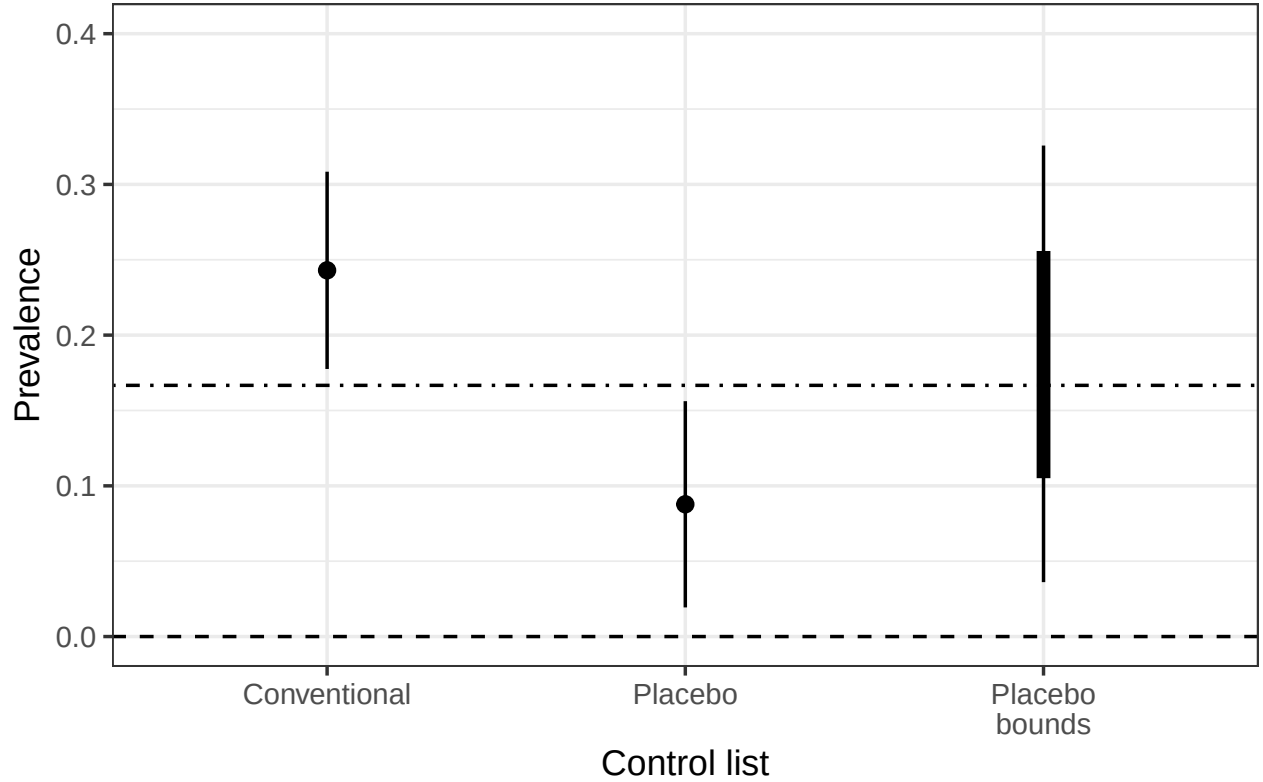


Figure 6: Prevalence estimates by type of control lists and known placebo bounds in Agerberg and Tannenbergs (2021)

Note: Thinner vertical lines denote 95% confidence intervals. Dot-dashed line indicates true prevalence of $1/6$.

bounds.¹² The bounds are estimated around the $[0.105, 0.256]$ range (CI range: $[0.036, 0.326]$). These bounds are valid, in the sense that they can be meaningfully interpreted as proportions, and informative in the sense that they contain the true prevalence and cover the range of the conventional estimate, suggesting that they are an accurate representation of the results that would have emerged in the absence of a placebo statement.

5 Conclusion

Placebo items can be used in list experiments to minimize potential bias from non-strategic errors. However, this puts the burden on choosing the right placebo item, since choosing one

¹²We could have also incorporated an estimator that uses covariate adjustment to further improve statistical precision to mimic the refinement in Section 3.3, but decided not to since we found no indication that any of the covariates available in the replication materials was meant to be used in this way.

with non-negligible prevalence, be it real or artificial, may attenuate prevalence estimates. In extreme cases, like the one we present, the bias introduced may even render estimates invalid.

We demonstrate how a partial identification approach may be used to recover informative estimates. Traditional sharp bounds alone are not sufficient to recover valid estimates, but a combination of assumptions justified by the list experiment design, auxiliary information about the prevalence of the placebo item in the population of interest, and using a more precise estimator can yield sufficiently informative bounds, which researchers can use to inform decision-making or subsequent iterations of a study.

This approach is useful even in contexts where researchers may already have individual-level information about placebo items. A concern in the literature with placebo-controlled list experiments is that true zero-prevalence placebo items (e.g. being abducted by aliens) can still be counted by respondents not because they are true but because they stand out, especially in online surveys. In this case, researchers can use the proposed approach to quantify a study’s sensitivity to potential issues with the choice of placebo items, which in turn would increase the applicability of this version of the list experiment.

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Data Availability Statement

Data, code, and materials needed to replicate this manuscript will be accessible via Harvard Dataverse.

Competing Interests

The authors declare none.

Research with Human Subjects

The research involving human subjects was reviewed and approved by the Research Ethics Committee at the University College Dublin [HS-C-22-115-Tiscornia].

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